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Research article

Dynamic hedging of 50ETF options using Proximal Policy Optimization

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ABSTRACT

This paper employs the PPO (Proximal Policy Optimization) algorithm to study the risk hedging problem of the Shanghai Stock Exchange (SSE) 50ETF options. First, the action and state spaces were designed based on the characteristics of the hedging task, and a reward function was developed according to the cost function of the options. Second, combining the concept of curriculum learning, the agent was guided to adopt a simulated-to-real learning approach for dynamic hedging tasks, reducing the learning difficulty and addressing the issue of insufficient option data. A dynamic hedging strategy for 50ETF options was constructed. Finally, numerical experiments demonstrate the superiority of the designed algorithm over traditional hedging strategies in terms of hedging effectiveness.

1. Introduction

Options are an important type of financial derivative, granting the holder the right to buy or sell an asset at a specific price on or before a certain date. Compared to other domestic financial products, options provide investors with a higher level of trading flexibility. In 2015, China's first listed option product, the 50ETF option, was launched on the Shanghai Stock Exchange. The underlying asset of this option is the SSE 50 Exchange Traded Fund (ETF). Since its introduction, numerous securities and futures firms have gradually opened stock options trading permissions, and the options market has experienced rapid growth.

The price of an option reflects the movement patterns of the option and its underlying asset in the real market. The earliest model for pricing European options, known as the Black–Scholes (BS) option pricing formula, was proposed by Fischer Black and Myron Scholes in 1973 [1], based on a series of assumptions. The "Greeks" of an option are metrics used to quantify various risk factors in an option position. By adjusting the Greeks of a portfolio, one can dynamically manage various risk exposures in the market. The most important Greek is Delta, and a perfectly dynamic hedge would maintain a Delta-neutral portfolio at all times. In reality, the assumption of continuous trading is not valid, as trading can only be done in discrete quantities, and each trade incurs costs (including fees and the bid–ask spread). This makes continuous Delta adjustment impractical, leading to the inability of Delta hedging based on the BSM model to achieve optimal hedging in real markets. To address these shortcomings of the BS model, many scholars have proposed a series of improvement techniques. For instance, in response to the issue of transaction costs not being considered in the BS model, Leland [2] constructed an adjustment factor proportional to transaction costs and incorporated this factor into volatility calculations, resulting in the Leland hedging model, which serves as a useful reference standard. For more information on work in this area, refer to [3,4].

Additionally, traditional option hedging methods often rely on experience and manual adjustments, lacking automation and intelligent features. Recently, deep learning and reinforcement learning have achieved significant results in various fields, including finance [5,6]. In recent years, deep reinforcement learning algorithms have begun to be applied in the field of option hedging, see [7,8]. These algorithms aim to optimize hedging by selecting actions at each step that maximize the total expected profit and minimize the variance of profits and losses, ensuring that the hedging performance achieved through the strategy is optimal. The literature [9] presents an option replication and hedging scheme based on reinforcement learning algorithms, which does not depend on traditional option pricing models, nor does it require detailed information such as strike prices of options. The literature [10] uses VaR and CVaR as two indicators to study and evaluate the risk levels of Delta, Delta_Gamma, Delta_Vega, and deep distributed reinforcement learning hedging strategies, finding that the probability of reinforcement learning methods experiencing greater risk is smaller. The literature [11,12] theoretically and experimentally illustrate the feasibility of applying reinforcement learning to option hedging, using Q-learning to construct a discrete-time option pricing and hedging model based on reinforcement learning. The literature [13] uses DQN

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and DDPG algorithms to implement an analysis of option hedging strategies for different market characteristics, transaction costs, option expiration dates, and hedging frequencies. In [14], the performances of DQN, DQN with Pop-Art, and PPO were compared for option hedging, revealing that PPO performed the best in terms of profit and loss, training time, and the amount of data required for training. Among the various deep reinforcement learning methods, the PPO algorithm stands out compared to other on-policy gradient methods, balancing the complexity, simplicity, and training time of samples. It also inherits the stability and reliability of the TRPO algorithm while being simpler and more general [15]. These advantages are quite important for dealing with financial issues. However, there is only a small amount of literature on the application of the PPO method to option risk control, so this paper uses the PPO algorithm for autonomous hedging based on simulated market data.

Inspired by the aforementioned literature, this paper employs the PPO method of deep reinforcement learning for the study of option hedging, which is expected to adapt to various market conditions and investment strategies, thereby enhancing the precision and efficiency of hedging. However, due to the relatively late development of the domestic options market, there are few relevant studies on option hedging. Additionally, the existing historical option data is insufficient to provide adequate training resources for deep reinforcement learning. In response, this paper employs the Black–Scholes model to generate simulated data for pre-training the model and incorporates the concept of curriculum learning, progressively increasing the difficulty of the learning tasks to bridge the gap between simulation and reality, thus optimizing the training process of the agent. Experimental results demonstrate that the proposed algorithm achieves effective hedging performance, confirming the feasibility and advantages of applying deep reinforcement learning algorithms to the SSE 50ETF options market.

2. Problem and model description

2.1. Problem description

In hedging, the aim is to reduce costs while simultaneously lowering risks. Assuming the agent has quadratic utility, the Mean–Variance model proposed by Markowitz [16] is considered.

$$\max\left(E[r_T]-\frac{k}{2}\mathrm{Var}[r_T]\right), \tag{1}$$

r_T represents the overall profit and loss of the entire investment portfolio, which encompasses both the market value changes at each time step and transaction costs. The risk emerges from the uncertainty surrounding the future value of the investment portfolio, characterized by the variance of the portfolio's value changes. To strike a balance between profit and risk, a risk aversion coefficient k is introduced.

$$W_T=\sum \Delta x_i \tag{2}$$

In this, the wealth increment at each moment already includes the market friction and transaction costs generated by buying and selling the underlying asset. For a transaction of n shares of the underlying asset, we define the transaction cost function as [17]:

$$C(n)=\text{transaction fee}+\text{market friction} \tag{3}$$

The trading cost consists of transaction fees and market friction, represented by the linear and quadratic terms with respect to n, respectively. In experiments, different values are assigned to the multiplier to simulate different trading cost scenarios.

2.2. The principle of reinforcement learning

The learning pattern of reinforcement learning agents is similar to that of humans. Within a period $(t=1,2,T)$, the agent needs to accomplish a certain task. Initially, the agent observes the environment and receives a state $s_t \in S$ from it. Based on the current state, it selects an action $a_t \in A$ to interact with the environment, which provides an immediate reward r_t. At this point, the environment transitions from state s_t to a new state s_{t+1}. Through continual interaction with the environment, the agent updates its actions in anticipation of obtaining higher rewards from the environment.

2.3. PPO algorithm framework

PPO (Proximal Policy Optimization) algorithm employs importance sampling techniques similar to those used in Q-learning to effectively reuse samples. It updates the target policy by sampling from the old policy. The improved policy gradient formula is as follows:

$$\begin{aligned}\nabla_\theta J(\theta)&=\mathbb{E}_{(s_t,a_t)\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t,\theta)Q_\pi(s_t,a_t)]\\&=\mathbb{E}_{(s_t,a_t)\sim\pi_{\theta_{\text{old}}}}[\frac{\pi_\theta(s_t,a_t)}{\pi_{\theta_{\text{old}}}(s_t,a_t)}\nabla_\theta \log \pi_\theta(a_t|s_t,\theta)\\&\qquad\times Q_{\pi_\theta}(s_t,a_t)]\end{aligned} \tag{4}$$

$Q_\pi(s_t,a_t)$ is the evaluation of taking a certain action in a state. To maintain consistency with the direction of the gradient $\nabla_\theta \log \pi_\theta(a_t|s_t,\theta)$, an advantage function has been introduced. Using this function instead of the action value function, it measures the quality of action a by assessing its advantage.

$$A_\pi(s,a)=Q_\pi(s,a)-V_\pi(s), \tag{5}$$

$$J(\theta)=\mathbb{E}(s_t,a_t)\sim\pi\theta_{\text{old}}\left[\frac{\pi_\theta(s_t,a_t)}{\pi_{\theta_{\text{old}}}(s_t,a_t)}A_{\pi_\theta}(s_t,a_t)\right]. \tag{6}$$

Simultaneously, in order to prevent instability in updates caused by significant discrepancies between the new and old policies, the ratio of the new policy to the old policy, denoted as $\frac{\pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)}$, is restricted within the range of $[1-\epsilon,1+\epsilon]$. Here, clip represents a truncation function, and ϵ stands for the truncation parameter:

$$\mathrm{clip}\left(\frac{\pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)},1-\epsilon,1+\epsilon\right). \tag{7}$$

Therefore, the objective function is adjusted to

$$\begin{aligned}L^{\mathrm{CLIP}}(\theta)=\hat{\mathbb{E}}_{(s,a)\sim\pi_{\theta_{\text{old}}}}&\left[\min\left(\frac{\pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)}\hat{A}_{\pi_\theta}(s,a),\right.\right.\\&\left.\left.\mathrm{clip}\left(\frac{\pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)},1-\epsilon,1+\epsilon\right)\hat{A}_{\pi_\theta}(s,a)\right)\right].\end{aligned} \tag{8}$$

This is the PPO-Clip algorithm with truncated terms.

3. Algorithm design

3.1. State–space design

Different state variables can have varying degrees of influence on the decision-making of the agent. This study considers the composition of the state space from the perspectives of options and underlying assets. Concerning options, the price C_t measures the value of the option, and the Delta(Δ_t) value serves as a significant risk metric that significantly affects the agent's policy choices. Looking at the underlying assets, the price S_t and the position size N_t are sources of value for the entire investment portfolio. Additionally, the duration of the option's maturity also influences its value. Other factors, such as the strike price K, remain constant throughout the lifecycle of the option, and the agent can easily capture such information in a short period. Therefore, these variables are not considered as state variables. The final state vector is represented as:

$$s_t := (t, S_t, C_t, \Delta_t, N_t).$$

Here, t denotes discretized hedge time stamps; S_t represents the price of the underlying liquid asset at time t; C_t represents the price of the option in the investment account at time t; Δ_t refers to the Delta value of the option in the investment account at time t (pre-hedging); N_t indicates the quantity of the underlying asset in the account at time t (pre-hedging).

3.2. Agent action design

The action space of the agent encompasses all possible instructions required to complete the task. In the realm of options hedging, defining the hedging task is necessary to specify the action space. Assuming a short position of 100 European call options, the action a_t chosen by the agent at each hedging time stamp is: the total quantity of underlying assets that the investment portfolio should hold before the next hedging time stamp arrives. This quantity represents the indirect hedging amount and is not directly used for hedging trades.

In order to improve the exploration efficiency of the agent, it is necessary to define the range of actions. Considering the Delta risk, the range of quantities of underlying assets needed for a short position of 100 call options should fall between 0 and 100. Hence, the action space for the agent is $[0, 100]$. However, having a continuous action space complicates the exploration for the agent and does not align with the integer trading rules in stock markets. Therefore, the original range of quantities for the underlying assets is discretized to integer values, which not only aligns better with trading actions in financial markets but also significantly reduces the dimensionality of the action space, thereby enhancing exploration efficiency. Ultimately, the agent's action space is defined as:

$$a_t \in \{0, 1, 2, \ldots, 100\}.$$

3.3. Reward function design

Trading costs refer to the costs incurred from buying and selling underlying assets, stemming from market frictions and transaction fees. For simplicity, transaction fees are calculated as a fixed proportion, denoted by α, of the traded amount of the underlying asset. Market frictions are determined by the concession made in terms of the price required for an immediate trade and are typically represented using a quadratic trading cost model. The cost function is expressed as follows:

$$f(S_t, \Delta N_t) = \alpha S_t \left(|\Delta N_t| + 0.01 \Delta N_t^2 \right). \tag{9}$$

At time t, the reward function is denoted as r_t. The price of the underlying asset is S_t, the quantity is N_t, the option price is C_t, and the bank account is B_t. Initially, there is a profit of $100C_0$, from shorting the call option and a long position with quantity N_0 of the underlying asset, incurring a cost of $N_0\ S_0$ that grows with a risk-free interest rate r. From time t to $t+1$ $(0 \le t < T)$, changes in the investment portfolio's value include: (1) the difference in the value of the underlying asset $N_{(t+1)}S_{(t+1)} - N_t S_t$; (2) the difference in the sold call option's value $100(C_{t+1} - C_t)$; (3) the trading costs for trading the underlying asset $(N_{(t+1)} - N_t)S_t$ and the associated fees and market frictions $f(S_t, \Delta N_t)$; (4) existing deposits or borrowings in the bank account B_t. The iterative formula for cash flow in the bank account is $B_{t+1} = B_t e^{rt\Delta t} - (N_{t+1} - N_t)S_t - f(S_t, \Delta N_t)$. The value of the entire investment portfolio at time t is $V_t = N_t S_t - 100C_t + B_t e^{rt\Delta t}$.

The total profit obtained by the entire investment portfolio at the expiration of the option is given by:

$$\begin{aligned}&(100C_0 - N_0 S_0)e^{rT} + (N_T S_T - N_0 S_0) - 100C_T \\ &\quad + \sum_{t=0}^{T-1} (B_{t+1} - B_t e^{r\Delta t}) e^{r(T-t)}.\end{aligned} \tag{10}$$

Equation (8) can be decomposed into profits at each time step. The wealth increment from time t to $t+1$ $(0 \le t < T)$ is represented as:

$$\begin{aligned}\Delta r_t &= (N_{t+1}S_{t+1} - N_t S_t) - 100(C_{t+1} - C_t) \\ &\quad - (N_{t+1} - N_t)S_t + f(S_t, \Delta N_t) e^{r(T-t)},\end{aligned} \tag{11}$$

Δr_t can be understood as the immediate return value for the agent at each step. The fluctuations in portfolio value originate from $\Delta r_t{}' = N_{t+1}(S_{t+1} - S_t) - 100(C_{t+1} - C_t) = N_{t+1}\Delta S_t - 100\Delta C_t$. Therefore, the variance of the final profit and loss of the investment portfolio is expressed as:

$$\mathrm{Var}(V_T') = \mathrm{Var}\left(\sum_{t=0}^{T-1} \Delta r_t' \right) \approx \sum_{t=0}^{T-1} \left((N_{t+1} - 100\Delta_t)\sigma S_t \right)^2 \Delta t, \tag{12}$$

where the σ is standard for the annualized volatility. The variance of profit and loss at step t is given by $\left[(N_{t+1} - 100\Delta_t)\sigma S_t\right]^2 \Delta t$. When the risk factor $k > 0$, the immediate reward generated at each step in every path is updated to the combination of wealth increment and the variance of profit and loss:

$$\begin{aligned}\Delta r_t =& (N_{t+1}S_{t+1} - N_t S_t) - 100(C_{t+1} - C_t) \\ &- \left((N_{t+1} - N_t)S_t + f(S_t, \Delta N_t) \right) e^{r(T-t)} \\ &- \frac{k}{2} \left[(N_{t+1} - 100\Delta_t)\sigma S_t \right]^2 \Delta t.\end{aligned} \tag{13}$$

The cumulative reward function for the agent at the end of each episode is obtained as:

$$\begin{aligned}R :=& E \Big[(100C_0 - N_0 S_0)e^{rT} + (N_T S_T - N_0 S_0) - 100C_T \\ &+ \sum_{t=0}^{T-1} \left(B_{t+1} - B_t e^{r\Delta t} \right) e^{r(T-t)} \Big] - \frac{k}{2} \mathrm{Var}[V_T'].\end{aligned} \tag{14}$$

3.4. PPO option hedging algorithm process design

PPO uses the PyTorch 1.13.1 deep learning framework. The PPO algorithm, based on the Actor–Critic framework, addresses issues related to continuous action spaces. The input layer of the Actor network receives a five-dimensional state from the environment feedback, and the network outputs parameters generating the probability distribution of actions. Similarly, the Critic network takes the same five-dimensional state as input and outputs the state value function, which plays a crucial role in computing the loss function for updating the parameters of both networks. Table 1 represents the algorithmic workflow of applying PPO to options hedging.

4. Empirical study

The experiment consists of two parts. Initially, simulated data is generated based on the Black–Scholes option pricing formula. The DQN and PPO algorithms are pre-trained using this simulated data. The objective is to obtain a foundational version of option hedging strategy at relatively low data cost. However, the models trained in this phase cannot be directly applied to the real market. Further fine-tuning of these existing models is necessary for the specific task of 50ETF option hedging. The ultimate goal is to derive an option hedging strategy based on the 50ETF options. The specific experimental steps are outlined in Fig. 1.

4.1. Experimental approach and data selection

The data used in this section consists of two parts: simulated data and real data. The simulated data is generated from the Black–Scholes (BS) model with an initial price S_0? and a fixed volatility σ. After pretraining, real option data from the WIND database is used for fine-tuning. The real data includes information from the China SSE 50ETF Fund (510050.OF) and its corresponding call option contracts, comprising the closing prices of the 50ETF fund from February 25, 2015, to August 31, 2022, as well as 44,640 records of call option contracts from September 2, 2021, to August 31, 2022. A portion of the options data is presented in Table 2.

4.2. Option hedging strategy design

(1) Reference model

The benchmark strategy for option hedging is set as Delta dynamic hedging, conducted once daily. At each hedging timestamp, the Delta value of the option in the portfolio is calculated, and the quantity of underlying assets is dynamically adjusted to ensure that the Delta value of the entire investment portfolio approaches or remains at zero. To serve as a comparative reinforcement learning model, the classic Deep Q-Network (DQN) algorithm, renowned for handling discrete action space problems, is chosen. DQN utilizes Tensorflow 2.9.2 as its deep learning framework.

(2) Evaluation metrics

The comparative approach in this section involves using the classic Delta dynamic hedging model as the baseline and the DQN algorithm as the benchmark model to assess the effectiveness of the PPO reinforcement learning algorithm in addressing option hedging problems within a simulated environment. The evaluation will be based on two categories of metrics:

(1) Cumulative reward and P&L (profit and loss) standard deviation: These metrics assess the quality of the reinforcement learning training process. Cumulative reward represents the accumulated rewards obtained by the agent in each round after each step of hedging transactions. Profit and loss standard deviation refers to the standard deviation of the investment portfolio's profit and loss for the most recent 30 rounds.

(2) Daily average profit and loss standard deviation, success rate, cumulative cost: These metrics evaluate the returns and risk conditions of each option contract after being processed by different hedging models. Daily average profit and loss standard deviation evaluates whether each hedging under the current strategy for the current option contract can generate relatively stable returns. Success rate indicates the percentage of positive gains obtained after transactions in each round. Cumulative cost is the sum of costs incurred in each round.

(3) Experimental Procedures

(1) Train the DQN and PPO algorithms using simulated data under scenarios of zero cost, low cost, and high cost. Adjust the algorithm parameters to optimize the performance of both models in each scenario. Compare the training effectiveness of the algorithms.

(2) Test the trained reinforcement learning models against the classic hedging method. Compare the hedging effectiveness of the models and summarize their strategy characteristics. Select the model demonstrating the best performance in the simulated environment.

(3) Organize and input 1700 real 50ETF option data into the best-performing pre-trained model. Conduct training optimization on this dataset.

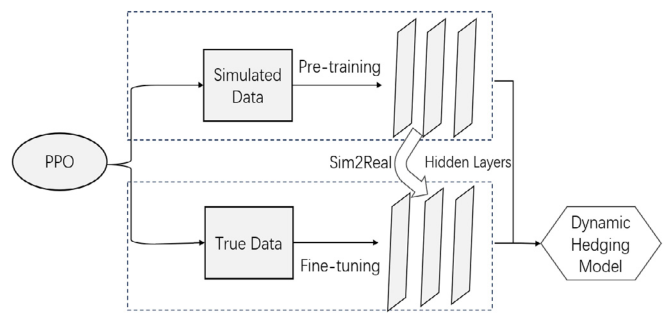


Fig. 1. Experimental design drawing.

Table 1
PPO algorithm flow.

		Algorithm flow: Option hedging strategy based on the PPO algorithm
1	Input:	State space $s_t := (t, S_t, C_t, \Delta_t, N_t)$
2	Output:	Quantity of underlying assets in the portfolio for the next time step N_{t+1}
3		Randomly initialize Actor network and Critic parameters θ, Ω
4	Initial Investment	Short 100 options, long N_0 underlying assets, and a bank account of $100C_0 - N_0S_0$
5	Repeat	The agent receives the initial state from the environment: $s_0 := (t_0, S_0, C_0, \Delta_0, N_0)$
6		Generate action N_{t+1} from the policy distribution produced by the Actor network
7		Obtain reward value $r_t = (N_{t+1}S_{t+1} - N_tS_t) - 100(C_{t+1} - C_t) - \left(N_{t+1} - N_t\right) S_t$ $+f(S_t, \Delta N_t)e^{r(T-t)} - \left(\left(N_{t+1} - 100\Delta_t\right)\sigma S_t\right)^2 \Delta t$
8		Update the quantity of underlying assets to N_{t+1} and obtain the new state s_{t+1}
9		Store sample trajectories$\{s_t, a_t, r_t\}$
10		End of the options cycle, reaching the final state s_T. At this point, the immediate return is the funds in the bank account $(100C_0 - N_0S_0)e^{rT}$
11		Calculate the estimated advantage function A_t based on the value function. Compute the ratio of new and old policies $r_t = \frac{\pi_\theta(a_t\|s_t)}{\pi_{\theta_{\text{old}}}(a_t\|s_t)}$ and obtain $L^{\text{clip}}(\theta)$
12		Update Actor network θ
13		Update Critic network Ω
14	End	

Table 2
Data information table of 50ETF options.

Date	Trading code	Exercise price(K)	Underlying closing price(S)	Days to maturity(t)	Closing price(C)	Delta
2019-09-02	510050C1909M02200	2.2	2.94	23	0.733	1
2019–09–02	510050C1909M02250	2.25	2.94	23	0.6814	1
…	…	…	…	…	…	…
2022–08–31	510050C2210M02950	2.95	2.801	56	0.0249	0.2373
2022–08–31	510050C2210M03000	3	2.801	56	0.0165	0.1699

(4) Set the benchmark model as the Delta dynamic hedging method with a 20% hedging band. Compare the performance of the benchmark model with the optimal hedging strategy on the real 50ETF option data in the test set.

4.3. Parameters setting

According to sections 3.1, 3.3, and 3.4 of this paper, a reinforcement learning environment is constructed, encompassing the configuration of the state space, action space, and reward function. In the case of DQN, discrete hedging actions can be directly obtained.

The neural network structure in the DQN is set to two hidden layers with 16 neurons each. To expedite computational speed, the activation function for the hidden layers is set to ReLU. The optimizer is set as Adam, with an initial learning rate of 0.001. Additionally, the Adam optimizer automatically adjusts the learning rates for different parameters during training. To assess the impact of reducing data noise on training stability, evaluation metrics such as cumulative reward and profit and loss standard deviation are used. The batch size is varied from 16 to 128. Moreover, while avoiding a slowdown in the learning rate of the agent, the experience replay buffer size is increased (10000, 50000, and 100000). This is to determine which data update cycle is more suitable for options hedging training. Finally, the optimal settings are determined as a batch size of 64 and an experience replay buffer size of 100000.

PPO is an algorithm designed for handling continuous action spaces. It requires restricting and rounding the output of the Actor network to obtain actions that meet the necessary conditions. For the PPO algorithm, both the Actor and Critic networks are configured with 2 hidden layers. The activation function for the hidden layers is set as ReLU. To minimize oscillations in gradient updates, both networks are trained using a constant learning rate SGD optimizer, with an initial learning rate of 0.001. Various learning rates within the range of 0.0001 to 0.1 were experimented with to strike a balance between convergence speed and training stability. It was determined that a learning rate of 0.02 effectively mitigates the fluctuation of the reward function. The batch size is set using a similar approach.

Another crucial parameter in the PPO algorithm is the clipping coefficient ϵ, used to limit the overall difference between old and new policies, aiming to reduce the oscillation amplitude of the reward function. The truncation parameter ϵ(0.1, 0.2, and 0.3) is adjusted towards achieving the best training outcome. The selection of relevant hyperparameters is as showed in Table 3.

Table 3
Hyperparameter settings of PPO.

Parameter name	Value
Actor network neuron count	[512,512]
Critic network neuron count	[512,512]
Batch size	32
Learning rate	0.02
Clipping parameter	0.2
Discount factor Gamma	0.99
Experience buffer CAPACITY	10 000
Act range	[0,100]

4.4. Model pre-training

(1) Zero-cost training

Fig. 2 depicts the training curves of the DQN and PPO algorithms under zero-cost conditions. The cumulative reward represents the accumulated return for each round, while the profit and loss standard deviation evaluate the fluctuation in model profits and losses. Observations indicate that both algorithms initially exhibit low cumulative rewards, ultimately converging near zero. However, DQN exhibits slower convergence, only starting to converge after 200 rounds. Both the cumulative reward curve and the profit and loss standard deviation curve indicate that PPO demonstrates better stability.

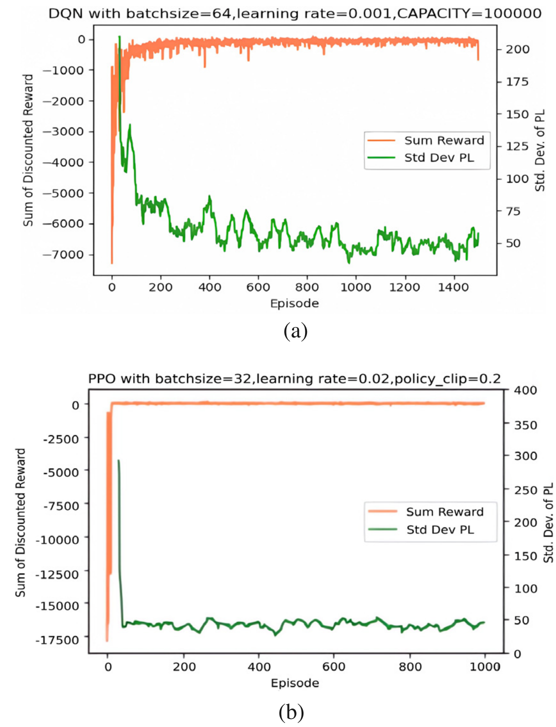


Fig. 2. Comparison of message delivery rates of three networks.

Fig. 3 illustrates the probability distributions of profits and losses for the Delta benchmark model, DQN, and PPO. Under zero-cost conditions, the profit and loss distributions of the three strategies approximate symmetry around zero. Both the DQN and PPO strategies attempt to increase opportunities for profit by appropriately exposing themselves to risks, resulting in more dispersed distributions. However, they yield different outcomes. As the round's profit and loss standard deviation increases, PPO exhibits higher average profits compared to the benchmark model. Conversely, DQN shows a severe fat-tail phenomenon on the left end, indicating the highest possibility of extreme negative returns. It could be inferred that DQN might struggle to effectively identify favorable risks during the hedging process.

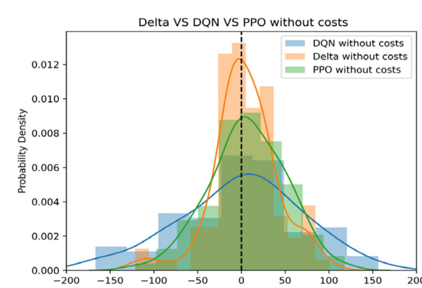


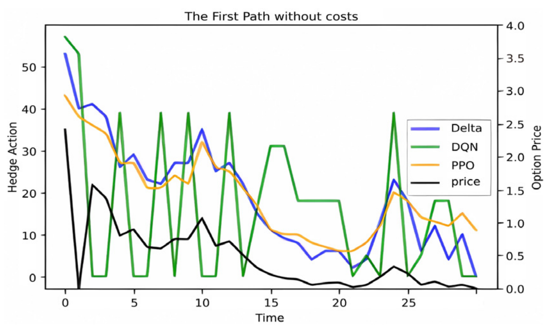
Fig. 3. Profit and loss distribution of different strategies in a no-cost environment.

Table 4 also presents the daily profit and loss standard deviation, average win rate, and average cumulative cost under the three strategies. The benchmark strategy exhibits the best performance, followed by PPO and then DQN. PPO manages to enhance returns by sacrificing a small amount of standard deviation. Under zero-cost conditions, all three strategies ensure a probability of portfolio value increase exceeding 0.5, and the benchmark strategy attains the highest average win rate.

Table 4
Results of model hedging in no-cost environment.

Model	Delta hedging	DQN	PPO
Average round profit (k)	4.29	−4.25	7.82
Round profit standard deviation	37.19	77	43.83
Daily profit and loss standard deviation	6.939	14.832	9.434
Average win rate per round (%)	66	59.3	62.03
Average cumulative cost (k)	0.0	0.0	0.0

Fig. 4 presents two hedging paths. All three strategies promptly adjust positions based on asset value changes. Compared to DQN, PPO's actions resemble Delta dynamic hedging and are more moderate. When there is no clear price trend information, PPO advocates hedging risk with a lower number of underlying asset holdings and adjusts actions incrementally based on small changes in trends. Conversely, DQN behaves entirely differently by making substantial changes in the quantity of underlying assets.



(a)

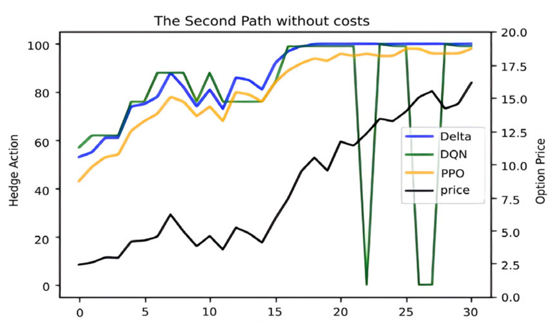


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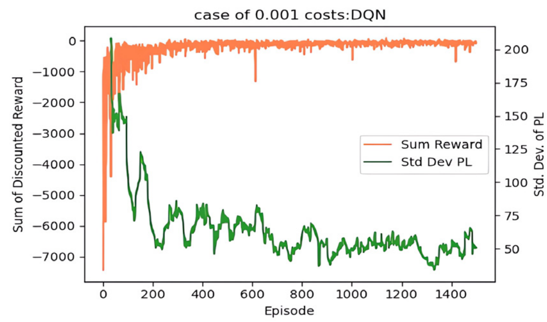
Fig. 4. Comparison of actions of different strategies in a no-cost environment.

(2) Low-cost training

In low-cost training, PPO continues to exhibit faster convergence speed and better stability, see Fig. 5. From the profit and loss probability distribution chart showed in Fig. 6, it can be observed that due to the presence of costs, the distributions of the three strategies have shifted to the left. Most profits and losses are concentrated between $[-60, 20]$. The kurtosis of the profit and loss distribution for the Delta dynamic hedging strategy is the highest. Both DQN and PPO successfully increase average profit by taking risks, but the distinction lies in DQN's strategy showing a fat-tail characteristic on the left side, indicating a higher probability of extreme low returns. Overall, PPO strategy incurs less additional risk while achieving higher average profits.

From Table 5, it is evident that the strength of the PPO strategy lies in reducing trading costs, successfully decreasing the volatility induced by costs. Therefore, the PPO strategy enhances profits without causing significant fluctuations in the portfolio value after each trading step within a round. In contrast, compared to the baseline model, the average cumulative cost of the DQN strategy increased by nearly double, and the increase in hedging costs did not improve the stability of profits and losses. Moreover, it can be concluded that incurring costs does not decrease the win rate of both reinforcement learning strategies.

When costs are involved, all three models can still replicate the trend of option prices but with slight adjustments in their actions, see Fig. 7. DQN continues to significantly adjust the quantity of the underlying asset with market fluctuations. However, for the most part, its holdings are lower than both the PPO strategy and the baseline strategy, while it compensates for the hedging costs by reducing the number of hedges. When the option trends towards being in-the-money, the PPO strategy almost replicates the actions of the baseline strategy.



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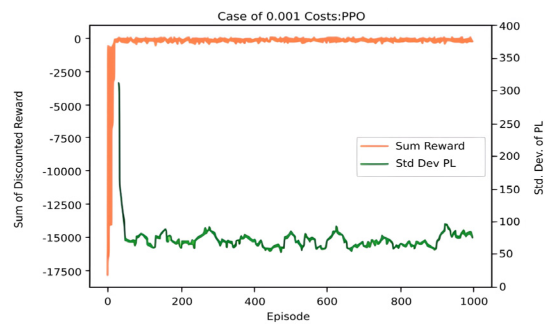

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Fig. 5. Comparison of training process of reinforcement learning algorithm in low-cost environment.

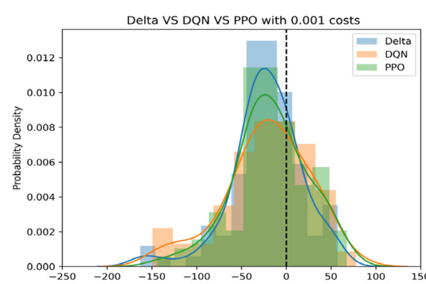
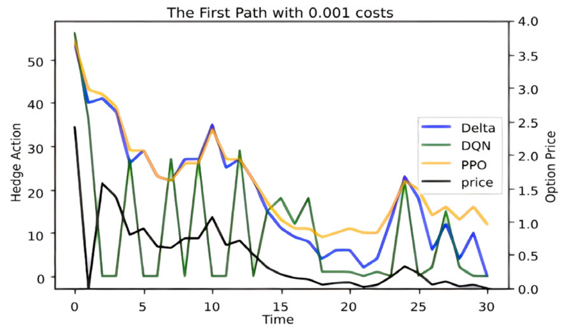


Fig. 6. Profit and loss distribution of different strategies in a low-cost environment.

Table 5
Results of model hedging in low-cost environment.

Model	Delta hedging	DQN	PPO
Average round profit (k)	−22.78	−21.69	−17.61
Round profit standard deviation	39.67	48.05	41.90
Daily profit and loss standard deviation	6.939	10.110	8.213
Average win rate per round (%)	65.76	63.83	62.63
Average cumulative cost (k)	0.368	0.711	0.293



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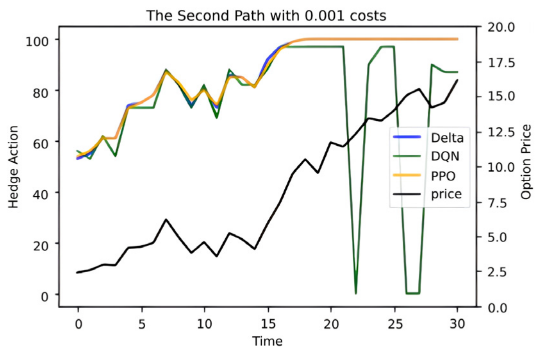

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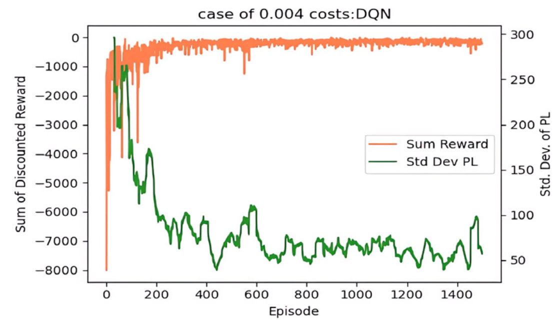
Fig. 7. Comparison of actions of different strategies in a low-cost environment.



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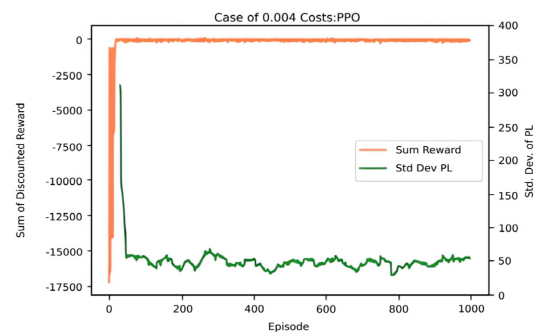

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Fig. 8. Comparison of reinforcement learning algorithm training process in high-cost environment.

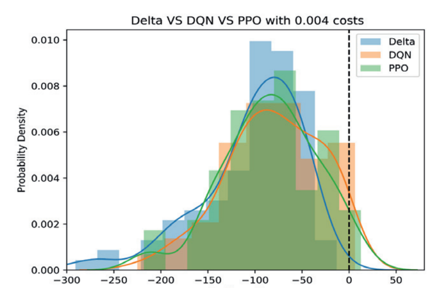

Fig. 9. Profit and loss distribution of different strategies in high cost environment.

At the same time, PPO adapts its actions in line with the overall trend of option prices, consistently adhering to the principle of making frequent small adjustments in hedging.

(3) High-cost training

In high-cost training, the PPO strategy maintains superior convergence and stability, see Fig. 8. On the test set, the gains of all three strategies are mainly negative, see Fig. 9. The difference between the baseline model and the two reinforcement learning strategies widens. The profit-loss distribution of Delta dynamic hedging shows a severe heavy-tail phenomenon on the left end. Although the average round-trip gains between DQN and PPO are similar, PPO shows significantly lower round-trip profit-loss standard deviation and daily average profit-loss standard deviation, see Table 6. This indicates that both inter-round and per-step profits are more stable with PPO.

In high-cost scenarios, compared to earlier conditions, the PPO strategy did not exhibit significant changes, see Fig. 10, indicating that PPO is not highly sensitive to variations in the cost ratio coefficient.

Table 6
Model hedging results in high cost environment.

Model	Delta hedging	DQN	PPO
Average round profit (k)	−102.61	−95.48	−96.47
Round profit standard deviation	53.65	49	44.51
Daily profit and loss standard deviation	6.939	11.136	8.945
Average win rate per round (%)	65.3	61.5	62.57
Average cumulative cost (k)	1.473	2.249	1.183

4.5. Empirical study of PPO on 50ETF options

In Section 4.4, continuing from the pre-trained models, the data for 50ETF call options contracts from September 2, 2019, to May 31, 2022, was used for further training. Directly feeding the 50ETF options data into the PPO pre-trained model yielded poor results, necessitating parameter adjustments to adapt to the new 50ETF dataset. Initially, experimenting with the crucial learning rate within the range of 0.0001 to 0.1, a learning rate of 0.0004 yielded optimal training results, as depicted in Fig. 11, eliminating the need for further parameter adjustments. The updated corresponding hyperparameter combinations are listed in Table 7. This process not only conserves real data resources but also rapidly achieves satisfactory results. Due to differing lifetimes of each contract, a continuous 30-day data segment was extracted from each contract as a single path.

In the real environment, the PPO algorithm initially explores learning towards increasing cumulative rewards and continually reduces the volatility of round profits. After 1000 rounds, the cumulative rewards gradually converge and eventually stabilize after 1500 rounds.

Fig. 12 depicts the probability distribution of gains and losses for the PPO strategy and the Delta dynamic hedging strategy with a 20% hedging band in the real environment of the 50ETF options. As the hedging strategy in this study does not involve directional judgments, the Delta dynamic hedging strategy exhibits a bimodal distribution, displaying significant differences in gains and losses between out-of-the-money and in-the-money options, clustered in distinct profit and loss regions. Conversely, the PPO strategy demonstrates greater stability, actively responding to price changes in various directions.

The contract 510050C2111M03400, a call option with a strike price of 3.4 and an expiration date of November 24, 2021, was randomly

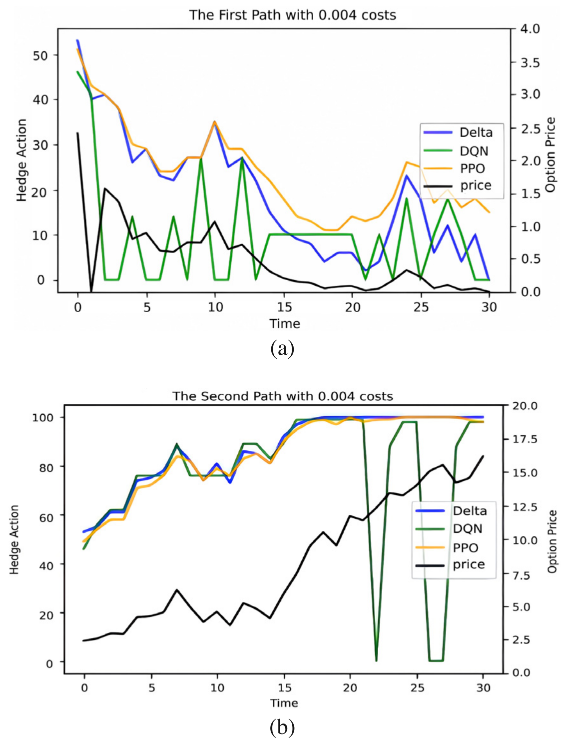

Fig. 10. Comparison of actions of different strategies in a high-cost environment.

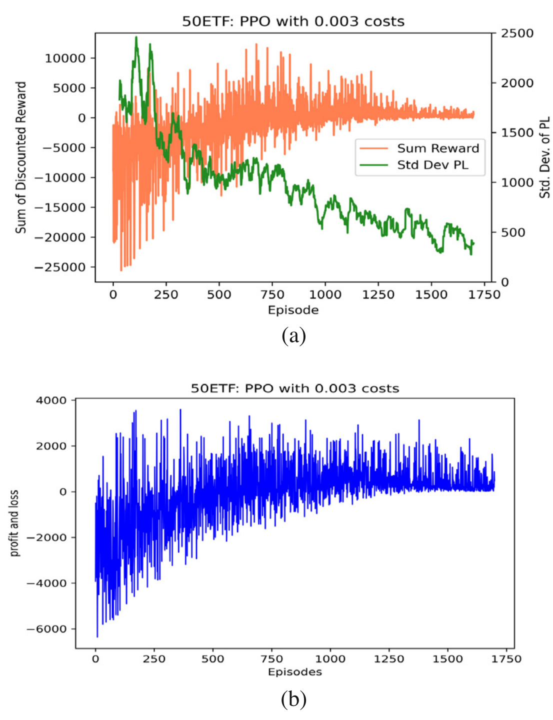

Fig. 11. PPO training process in real environment.

selected for validation. A comparison of hedging transactions was conducted over 30 trading days from October 12, 2021, to November 23, 2021 (see Table 8).

Fig. 13 illustrates that during the initial 8 days, the underlying asset price increased while the portfolio value decreased. From the 9th day onwards, as the underlying asset price declined, the PPO strategy opted for partial hedging to increase risk return. Towards the later period, the option became significantly out-of-the-money, and PPO tended to stop hedging, whereas Delta consistently maintained the highest position. Compared to the benchmark strategy, the PPO strategy achieved more

Table 7
Hyperparameter Settings of PPO in 50ETF environment.

Parameter name	Value
Actor network neuron count	[512,512]
Critic network neuron count	[512,512]
batch size	32
learning rate	0.0004
Clipping parameter	0.2
Discount factor Gamma	0.99
Experience buffer CAPACITY	10 000
act range	{0,1,2, …,10000}

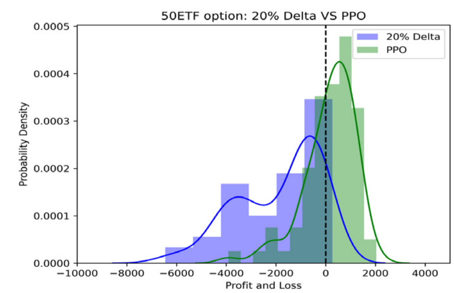

Fig. 12. Profit and loss distribution of PPO and benchmark strategies.

Table 8
Results of model hedging in real environment.

Model	20%Delta	PPO
Average daily profit and loss(k)	−3.754	24.512
Daily P&L standard deviation	84.833	126.888
win ratio(%)	53.333	63.333
Cumulative cost(k)	237.832	40.501

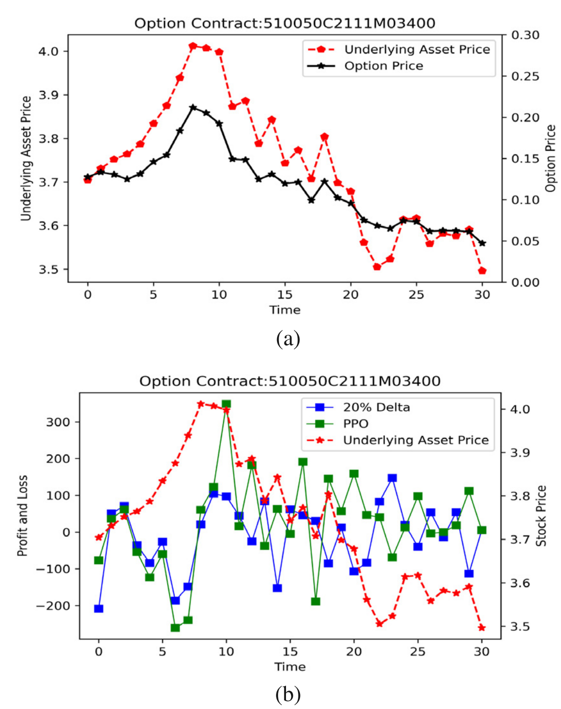

Fig. 13. Price paths and comparison of actions of different models in real environments.

positive returns, saving a considerable amount of costs, albeit inevitably increasing the volatility of returns.

5. Conclusion

This study explores the risk hedging problem of SSE 50ETF options based on the PPO algorithm. By designing action and state spaces suited to the hedging task and constructing a reward function based on the option cost function, a dynamic hedging strategy was proposed. To address the issue of insufficient option data, the research adopted the concept of curriculum learning, guiding the agent to progressively learn from simulated to real environments, thereby reducing task difficulty. The results of numerical experiments demonstrate that the proposed algorithm outperforms traditional hedging strategies in terms of hedging effectiveness, showcasing strong practicality and efficiency.

CRediT authorship contribution statement

Lei Liu: Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Mengmeng Hao:** Writing – original draft, Software, Data curation. **Jinde Cao:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lei liu reports financial support was provided by Hohai University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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